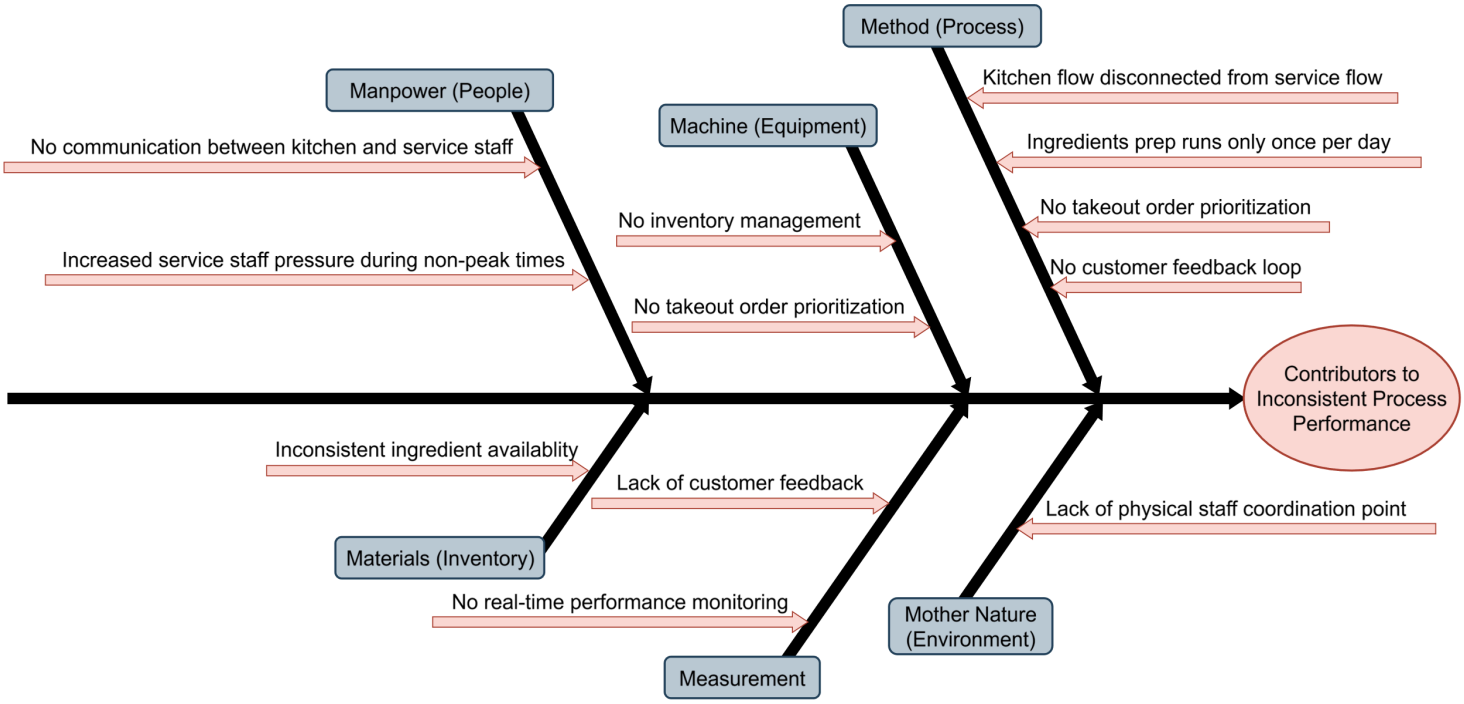
A. Udon-ya Process Improvement Solution

Udon-ya's primary objective is to establish an efficient and standardized order process that can be consistently reproduced and utilized by future franchisees. Currently, Udon-ya's process is suffering from inefficiencies and inconsistent performance.

Root cause analysis using the 6M framework presented by Yankelevitch (2022) identified contributing factors across six categories: disconnected kitchen and service workflows (Method), communication gaps across stakeholders (Manpower), lack of inventory management (Machine), inconsistent ingredient availability (Materials), absence of performance monitoring (Measurement), and lack of physical coordination points (Mother Nature/Environment). These six categories, or the main branches in the following fishbone diagram, represent bottleneck areas in Udon-ya's current order process:



The process mining analysis in Celonis of 9,713 cases confirmed 1,094 process variants and a conformance rate of 35.11%, highlighting significant variability in current process execution.

Throughput time analysis revealed bottlenecks in ingredient preparation, order fulfillment, and service coordination. These bottlenecks collectively contribute to inconsistent performance and represent opportunities for process improvement.

A2. Process Improvement Methodology

Lean methodology will be applied to address the identified bottlenecks. Lean focuses on waste elimination and process standardization, aligning with Udon-ya's goals for franchise expansion.

The six identified bottlenecks manifest as specific Lean waste types. Disconnected kitchen and service workflows (Method) create waiting waste where activities do not proceed without coordination. Communication gaps across stakeholders (Manpower) and a lack of physical coordination points (Environment) result in motion waste due to unnecessary coordination efforts. Lack of inventory management (Machine) and inconsistent ingredient availability (Materials) create inventory waste through both excess preparation and ingredient shortages. Absence of performance monitoring (Measurement) creates underutilized talent waste by failing to capture operational insights.

Improvement 1: Process Integration

Continuous flow is a Lean principle where activities progress sequentially through steps, adding incremental value at each stage, without waste (Lean Enterprise Institute, 2025). This improvement creates continuous flow by integrating previously separated kitchen and service workflows through two coordination activities: Update Ingredients Kanban (mutual updates where service signals demand and kitchen signals availability) and Check Ingredients Level (service monitors demand).

These new activities create communication loops that connect kitchen activities to service delivery, and the Ingredients Kanban board serves as the centralized coordination point. This improvement addresses three bottlenecks: disconnected workflows (Method), communication gaps (Manpower), and a lack of physical coordination points (Environment) by eliminating waiting waste in the form of idle time between kitchen and service activities, as well as motion waste in the form of uncoordinated communications.

Improvement 2: Inventory Pull System

Pull systems are a Lean principle for inventory management where production is triggered by actual customer demand rather than forecasts (Lean Enterprise Institute, 2025). This improvement implements a pull system using the kanban board to trigger ingredient preparation based on real-time inventory depletion rather than a fixed single batch.

The pull system operates through decision gateways that check kanban status and trigger corresponding preparation cycles only when ingredients are low. Kitchen preparation activities respond to these signals, with Deliver Prepped Ingredients completing the cycle by replenishing kanban inventory.

This improvement addresses two bottlenecks: lack of inventory management (Machine) and inconsistent ingredient availability (Materials) by eliminating inventory waste and associated delays through customer demand-based preparation cycles.

Improvement 3: Takeout Order Prioritization

Standardization is a Lean principle that creates consistent, repeatable processes by establishing uniform procedures for routine tasks (Lean Enterprise Institute, 2025). This improvement implements standardized order routing through the Takeout decision gateway and Order Prioritization activity, which ensures consistent handling of dine-in versus takeout orders.

The automated system classifies each order upon entry through the Takeout gateway, routing takeout orders through the Order Prioritization activity for systematic handling. In contrast, dine-in orders proceed directly to the "Customer Seated" activity, eliminating manual routing decisions and ensuring that takeout orders receive the appropriate priority in the service workflow.

This improvement addresses the lack of an order prioritization system (Machine) bottleneck by eliminating waiting waste in the form of customer delays at the customer entrance and order delivery through consistent, automated order classification and routing.

Improvement 4: Continuous Improvement

Continuous improvement (Kaizen) is a Lean principle that focuses on ongoing improvements driven by the collection and analysis of feedback (Lean Enterprise Institute, 2025). This improvement implements a feedback loop through the 'Customer Gives Feedback' activity, systematically collecting and analyzing feedback for process improvement. This data will enable Udon-ya to monitor customer satisfaction as a process quality metric.

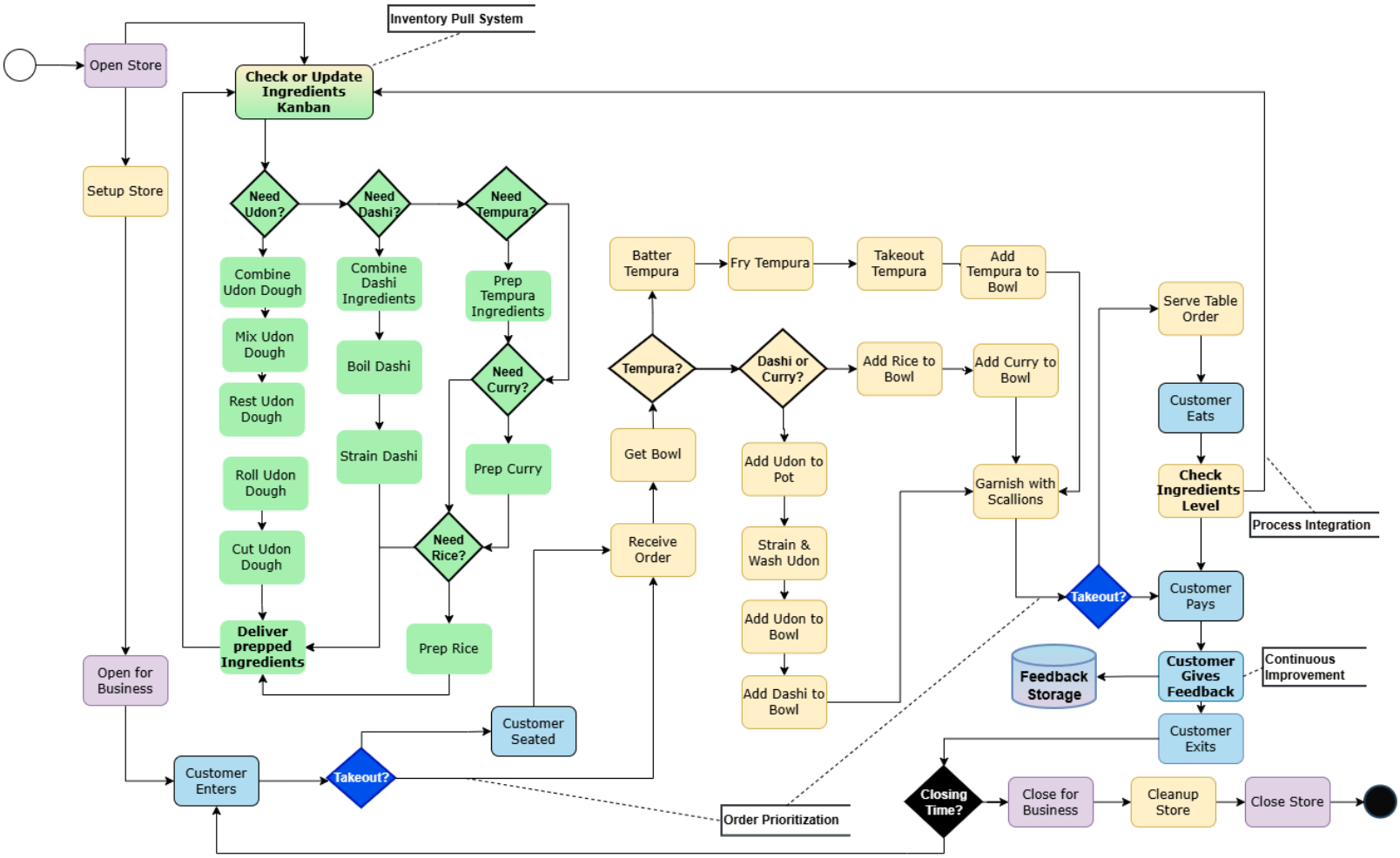
This improvement addresses the absence of performance monitoring (Measurement) by eliminating underutilized talent waste in the form of uncaptured customer and staff insights, enabling data-driven process refinement and supporting franchise standardization.

Lean Solution and Udon-ya Business Needs

Together, these four Lean improvements create continuous flow throughout Udon-ya's order process while also addressing all six identified bottlenecks. Process integration and pull systems eliminate the singular batch ingredients preparation cycle that created disconnected operations. Automated takeout order classification ensures an appropriate customer experience, and continuous improvement feedback ensures ongoing optimization.

This integrated Lean approach creates a standardized, repeatable process essential for franchise application, while eliminating the waste types that caused the 35.11% conformance rate and 1,094 process variants revealed in the Celonis analysis. By systematically addressing waiting waste, motion waste, inventory waste, and underutilized talent waste, this solution enables Udon-ya to achieve consistent process performance suitable for successful franchise expansion.

A3. Improved Order Process Flowchart



The process flowchart above illustrates the improved order process, which incorporates the four Lean improvements. The color coding represents different process participants: green activities indicate kitchen operations, tan activities indicate service operations, light blue activities indicate customer touchpoints, and purple activities relate to management responsibilities.

Process integration introduces two new coordination activities: 'Update Ingredients' Kanban, performed by both the kitchen and service staff, and 'Check Ingredients Level' for service staff. These activities create communication loops between kitchen and service workflows, replacing the previously disconnected operations where kitchen and service functioned as separate processes with no coordination.

The inventory pull system operates through decision gateways (Need Udon?, Need Dashi?, Need Tempura?, Need Curry?, Need Rice?) that check kanban status and trigger corresponding kitchen preparation activities only when ingredients are low. Multiple on-demand ingredients preparation cycles replace the current single daily batch cycle to maintain optimal inventory levels throughout operations. Takeout Order Prioritization routes orders through the Takeout decision gateway (blue diamonds) and automates the Order Prioritization activity, ensuring systematic classification and handling of takeout versus dine-in orders.

The Customer Gives Feedback activity automates the collection of customer feedback and stores the data in Feedback Storage for analysis. This feedback loop replaces the current absence of feedback collection and metrics with a systematic way to capture, retain, and analyze customer insights that inform ongoing process optimization.

A4. Implementation Plan

● Implementation Strategy

Based on the interdependent nature of the improvements, implementation will follow a two-phased approach that prioritizes coordination changes in the first phase, and the remaining enhancements in the second phase. The total timeline for all changes is six weeks.

Phase 1: Foundation (Weeks 1-4) - Process Integration and Inventory Pull System
Udon-ya should implement process integration and inventory pull system improvements concurrently, as the kanban board serves as both the physical integration point and the pull system signal, making them inseparable components of the solution.

Roles: Management, Kitchen Staff, Service Staff

Task/Deliverable	Timeline	Responsible	Consulted
Install Kanban board	Week 1	Management	Kitchen/Service
Define "low ingredients" threshold	Week 1	Management	Service
Kitchen staff training	Week 2	Management	Kitchen
Service staff training	Week 2	Management	Service
Test process improvements	Week 3	Kitchen/Service	Management
Go live with integrated workflow	Week 4	Management/Kitchen/Service	

Resources: Kanban board materials, training time (4 hours per staff member), process documentation

Phase 2: Enhancement (Weeks 5-6) - Takeout Order Prioritization and Customer Feedback
Once the integrated workflow is stable, add automated takeout order prioritization and customer feedback collection.

Roles: POS Vendor (system configuration), Management, Kitchen Staff, Service Staff

Task/Deliverable	Timeline	Responsible	Consulted
Configure the POS system for takeout routing and collecting customer feedback.	Week 5	Management	POS Vendor
Set up customer feedback data storage	Week 5	Management	POS Vendor
Test system changes	Week 5	Management	
Train service staff on system changes	Week 6	Management	Kitchen/Service
Launch system changes	Week 6	Management/Kitchen/Service	

Resources: POS vendor support (8 hours), training time (15 mins per staff member)

- **Risk Management Plan**

Risk	Likelihood	Impact	Severity	Mitigation Strategy
Kitchen staff resistance to frequent ingredients prep cycles instead of once per day	Medium	**High**	**High**	Involve kitchen staff in setting kanban thresholds and designing the kanban methodology.
Service staff resistance to monitoring process adherence	Medium	Medium	Medium	Emphasize the time savings from Kanban versus asking kitchen staff, as well as better availability of ingredients on high-volume days.
Kanban signals missed	**High**	Medium	**High**	Place the kanban board in a place that is easily visible to both the kitchen and service staff at all times.
POS may not be able to accommodate customer feedback collection	Low	**High**	Medium	Explore alternatives for consistently collecting customer satisfaction data.
POS vendor delays	Low	**High**	Medium	Engage the POS system vendor early, in Week 3 of Phase 1
An inaccurate low ingredients threshold can cause over- or under-preparation.	Medium	Medium	Medium	Monitor closely during Week 4 go-live. Adjust thresholds based on observed demand data in week 4. Repeat until ideal.

- **Change Management Considerations**

To implement these Lean improvements, Udon-ya requires systematic change management to ensure staff adoption and minimize resistance.

Managing Organizational Change:
The two-phased implementation approach serves as a change management strategy by introducing changes incrementally rather than overwhelming staff with simultaneous modifications. Phase 1 establishes the foundational coordination system, allowing staff to adapt to integrated workflows before

adding system enhancements in Phase 2. This approach allows for time to learn, adjust, and build confidence with each improvement before introducing the next.

Management will maintain visible involvement throughout the implementation by participating in daily operations during testing, demonstrating commitment to the changes, and providing support when staff encounter challenges. Regular check-ins during Week 4 go-live and Week 6 enhancements launch create opportunities for staff to voice concerns and suggest refinements, which facilitates a collaborative culture.

Strategy for Overcoming Resistance:
Early involvement and transparent communication about changes and benefits help overcome resistance. Kitchen and service staff will participate in defining kanban thresholds and board placement during Week 1, thereby creating ownership of the coordination system and transitioning staff from passive recipients of change to active contributors in designing the solution.

Clear communication of personal benefits helps overcome resistance by showing staff how changes reduce their current frustrations. Kitchen staff will understand that demand-based preparation eliminates both ingredient shortages during rushes and wasted prep during slow periods, while service staff will recognize that the kanban board eliminates time spent searching for kitchen staff to check

ingredient availability. Framing changes as solutions to existing pain points rather than additional work requirements increases acceptance.

Celebration of early wins reinforces positive adoption. When the first successful shift operates without ingredient shortages or when service staff complete orders more efficiently using the kanban system, management should publicly acknowledge these successes. Staff who effectively use new coordination activities will be recognized, creating positive examples for others.

A5. Training Plan

Training Session	Objectives	Methods	Schedule
Phase 1 – Kitchen Staff	● Understand kanban signals ● Updating kanban status ● Using decision gateways	● Hands-on kanban demo ● Practice scenarios with decision gateways ● Role-play with service staff	Week 2 1-2 hours Pre-shift
Phase 1 – Service Staff	● Monitor ingredient levels ● Updating kanban status	● Hands-on kanban demo ● Practice scenarios with combinations of ingredients ● Role-play with kitchen staff	Week 2 1-2 hours Pre-shift
Phase 1 – Cross-training	● Understand upstream and downstream impacts of each team's workflow ● Foster teamwork and ownership	● Full role-play with both teams ● Discuss the importance of teamwork ● Incentivize communication about process issues or inefficiencies	Week 3 1-2 hours Pre-shift
Phase 2 – Management	● Access feedback data	● Hands-on demo from the POS system vendor	Week 6 1 hour During operations
Phase 2 – Service Staff	● Ask for takeout status first ● Understand the feedback review process	● POS system takeout demo ● Practice transactions with reviews	Week 6 1 hour Pre-shift

A6. Communications Plan

Stakeholder	Key Messages	Communications Timing and Channels
Kitchen Staff	● Kanban eliminates ingredient shortages ● Prepping based on demand is less wasteful of efforts ● Staff input required for process success	Week 1: Pre-implementation meeting Week 1-2: Visual process map and instructions near Kanban Week 2: Daily meetings during process testing Week 3+: Ongoing staff meetings
Service Staff	**Phase 1:** ● No more searching for kitchen staff for more ingredients ● Faster order delivery ● Simple kanban update process	**Phase 1:** Week 1: Pre-implementation meeting Week 1-2: Visual process map and instructions near Kanban Week 2: Daily meetings during process testing Week 3-4+: Ongoing staff meetings **Phase 2:**

	Phase 2: ● Takeout routing improves customer experience ● Customer feedback helps with continuous improvement ● One-button takeout process replaces the need to ask if the order is takeout more than once	Week 5: Pre-implementation meeting Week 6: Training Week 6+: Ongoing staff meetings

A7. Executive Summary

Key Findings:

Udon-ya's order fulfillment process suffers from inconsistencies that threaten the viability of franchise expansion. Process mining analysis revealed 1,094 process variants, with only 35.11% conformance to a standard process, primarily due to six interconnected bottlenecks: disconnected kitchen-service workflows, communication gaps, inadequate inventory management, inconsistent ingredient availability, a lack of performance monitoring, and insufficient physical coordination points.

Recommendations:

This proposal recommends four integrated Lean improvements to address these bottlenecks: Process Integration creates continuous flow through new coordination activities; Inventory Pull System implements demand-based preparation; Automated Takeout Order Prioritization standardizes order routing; and Continuous Improvement enables systematic feedback collection. These improvements eliminate waiting waste, motion waste, inventory waste, and underutilized talent waste, which cause current metric failures.

Benefits for Udon-ya:

A two-phase, 6-week implementation establishes coordination systems before adding enhancements, with comprehensive risk and change management strategies to ensure staff adoption. This solution creates the standardized, repeatable processes essential for successful franchise replication and expansion.

B. Key Performance Indicators (KPIs)

KPI 1: Process Conformance Rate (new)

- Process Conformance Rate measures the percentage of order fulfillment cases that follow the standard integrated process without deviations. This KPI directly indicates the success of standardization, which is essential for franchise replication.

- Data will be collected monthly by Management by uploading POS timestamp data into the Celonis process mining system, using automated analysis that compares actual process execution to the process model. Udon-ya will analyze the number of unique variants and the most common process deviation patterns to identify areas requiring correction.

- **Target Value:** >80% (baseline: 35.11%)

- **Justification**: This KPI is suitable for evaluating process improvement success because it directly measures the reduction in process variation that led to the initial inconsistency problem. Increasing conformance from 35.11% to >80% demonstrates that the Lean improvements have created the standardized, repeatable processes necessary for franchise expansion.

KPI 2: Food Preparation Time (existing)

- Food Preparation Time measures the average time from receipt of the order to when the food is ready to serve. This KPI assesses the effectiveness of the pull system in eliminating ingredient delays.

- Data will be collected daily from POS timestamp data by Management, and uploaded to Celonis to calculate the time difference between order entry and meal completion timestamps. All orders are measured continuously.

- **Target Value**: <5 minutes consistently (current: exceeds 5 minutes during overall high volume)

- **Justification**: This KPI is appropriate because it directly measures whether the Inventory Pull System eliminates the ingredient shortage delays identified in the bottleneck analysis. Consistently achieving the 5-minute target across all volume periods indicates successful demand-based preparation.

KPI 3: Ingredients Replenishment Frequency (new)

- Ingredients Replenishment Frequency measures the number of ingredient preparation cycles triggered per day through the pull system. This KPI validates that the pull system is responding to actual demand rather than operating as a single batch.

- Data will be collected daily from kanban board logs by Kitchen Staff, recording each time a decision gateway triggers a preparation cycle.

- **Target Value**: 8-12 cycles per day (baseline: 1 batch per day)

- **Justification**: This KPI is appropriate because it directly measures whether the pull system operates as designed, triggering multiple demand-based preparation cycles throughout the day, rather than the once-daily batch approach. The shift from 1 to 8-12 cycles demonstrates demand-driven preparation that maintains optimal inventory levels and validates the elimination of inventory waste.

KPI 4: Average Time from Order to Table (existing)

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- Average Time from Order to Table measures the end-to-end time from when the restaurant receives the order until food reaches the customer. This KPI assesses the overall effectiveness of continuous flow across the integrated kitchen-service workflow.

- Data will be collected daily from the POS system data by Management, calculating the time difference between order receipt and order delivery timestamps. All orders are measured continuously.

- **Target Value:** <7 minutes during all volume periods (current: over target for high volume days)

- **Justification**: This KPI is appropriate because it measures the combined impact of all four Lean improvements on overall process performance. Consistently achieving the 7-minute target demonstrates that process integration and continuous flow have eliminated the waiting and motion wastes that previously caused delays. This metric also directly impacts customer experience, making it essential for franchise operations.

KPI 5: Customer Satisfaction Score (CSAT)

- CSAT measures the average rating customers provide on their order experience. This KPI assesses whether the Lean improvements effectively enhance the customer experience through faster service, consistent quality, and reliable ingredient availability.

- Data will be collected weekly from the Feedback Storage database (or POS system) by Management, calculating the average satisfaction rating across all feedback submissions. The POS system prompts customers to rate their experience on a 5-point scale at the point of payment.

- **Target Value**: Average rating ≥4.0 out of 5.0

- **Justification**: This KPI is appropriate because it directly measures the impact of process improvements on customers. Improved satisfaction scores validate that eliminating waiting waste, motion waste, and inventory waste translates to a better customer experience. This metric also provides data for the continuous improvement feedback loop, enabling ongoing process refinement from the customer's perspective.

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